Four-dimensional parallel insertions: one-loop shifts and the logarithm

One directional cut in four dimensions has the six coupling shifts (4) and (10) below, as explicit, convergent Brillouin-zone integrals. They extend the $SU(2)$ mid-plane calculation with its incorporated corrections to three coupled transverse planes, arbitrary positive anisotropy and cut fraction s. The bridge mass makes every one-step integral regular at zero momentum.

The universal logarithm belongs to the matched, four-dimensional background-field determinant. Its coefficient is analytically

$$2b_0 = \frac{11N}{24\pi^2}, \qquad g_0^{-2}(2a) - g_0^{-2}(a) = -\frac{11N}{24\pi^2}\log 2 + O(g_0^2) \tag{1}$$

when the endpoints use the same coupling scheme. For a full blocking step between different action shapes, the right side additionally contains $c_{\mathrm{in}} - c_{\mathrm{out}}$, equation (15). Returning the six heat times and the anisotropy to their initial values leaves that difference undetermined. The exact pushforward generates interactions already at tree level; carrying them through the next elimination is essential. Consequently four applications of the unperturbed one-step coefficients alone do not establish (1). The missing task is matching along the full generated-action trajectory, or a stationary action family under rescaled blocking. The composition treatment supplies a formal finite matching change for one pristine step and states the further depth-uniform requirements. Numerical evaluation of one lattice integral would leave this obstruction in place.

Status and inputs. Theorem 1 concerns coefficients of a formal, zero-image, bulk Laplace expansion. Its matrix and integral identities are written derivations; uniform remainder estimates remain open. Theorem 2 is a conditional one-loop matching theorem, with the continuum logarithm derived analytically. Neither theorem establishes a continuum measure or a physical mass gap. Inputs read here: the series/parallel note, Proposition 1, heat-time table, Hypothesis P(α), including its size clause, and §5; the mid-plane calculation, including N2, N3 and the winding qualification; the midpoint formulas, Theorems 1–4; and the zero-spacing note, perturbative column and finite Λ matching (full-read: these sections). These are internal written derivations, with no independent peer-review claim.

1. Normalization and a positive mid-space Hessian

Set $t = g^2 = \hbar g_{\mathrm{cl}}^2$ and

$$\tau_{\mu\nu} = \frac{a_\mu a_\nu}{\prod_{\rho\neq\mu,\nu} a_\rho}, \qquad t_{\mu\nu} = t\tau_{\mu\nu}.$$

Cut direction 1 at $s \in (0,1)$. With the trapezoid dual-length weights of atlas §1b, every old and new transverse face has heat time $2t\tau_{jk}$, $j,k \in \{2,3,4\}$. The cut faces have times $st\tau_{1j}$ and $(1-s)t\tau_{1j}$. This prescription extends to $D=4$ because only the dual length in direction 1 changes. It describes one cut of a uniform anisotropic lattice.

Write $m_e = m_{*e}e^{\xi_e}$ in the mid-vertex gauge, put $\sigma = s(1-s)$, and use dimensionless momenta $k \in \mathcal{B}_3 = [-\pi,\pi]^3$ with measure $d^3k/(2\pi)^3$. Define the three-by-three curl symbol and positive diagonal matrices by

$$\begin{aligned} (C\xi)_{jk} &= (1-e^{ik_k})\xi_j + (e^{ik_j}-1)\xi_k, \quad j<k, \\ M_{jj} = M_j &= \frac{1}{\sigma\tau_{1j}}, \qquad W_{jk,jk} = W_{jk} = \frac{1}{2\tau_{jk}}, \\ H_0(k) &= M + C(k)^* W C(k), \qquad G(k) = H_0(k)^{-1}. \end{aligned} \tag{2}$$

The exponent at zero coarse field is $\langle \xi, H_0\xi\rangle/(2t)$, so the covariance is tG. There is an identity in colour space. For fixed positive ratios,

$$M_{\min} I \le H_0(k) \le (M_{\max} + 12 W_{\max}) I. \tag{3}$$

Indeed C^*C has eigenvalues $0, Q, Q$, where $Q = \sum_{j=2}^{4}(2 - 2\cos k_j) \le 12$. In particular $H_0(0) = M$. For isotropic midpoint cutting, $H_0 = 4I + C^*C/2$ and $4I \le H_0 \le 10I$. The three-dimensional conditional theory has a bridge mass in lattice units, even as $t \downarrow 0$.

We extract the coefficient of $|X_{\mu\nu}|^2$ in the bulk local amplitude A_2 of the normalized defect $\mathcal{D}$. If that coefficient is $A_{\mu\nu}$ per coarse plaquette, write

$$\Delta(t_{\mu\nu}^{-1}) = 2A_{\mu\nu} + O(t), \qquad \delta_t^{\mu\nu} = 2t\tau_{\mu\nu}A_{\mu\nu} + O(t^2).$$

Thus the shift is $O(t)$ **relative** to the inverse heat time; its absolute leading value is $O(1)$. In continuum units the inverse coupling shift for this plane is $2\tau_{\mu\nu}A_{\mu\nu}$, since $a_\mu^2 a_\nu^2 / \prod_\rho a_\rho = \tau_{\mu\nu}$. All statements about $O(t)$ below refer to formal loop order at fixed heat-time ratios.

2. N2: the three cut-plane shifts

Theorem 1(a). In the formal bulk expansion, for $SU(N)$ with $C_A = C_2(\mathrm{adj}) = N$ and the metric of the input notes,

$$\boxed{\begin{aligned} A_{1j} &= \frac{C_A\sigma}{24}\int_{\mathcal{B}_3}[1 - M_j G_{jj}(k)], \\ \delta_t^{1j} &= \frac{C_A\sigma\, t\tau_{1j}}{12}\int_{\mathcal{B}_3}[1 - M_j G_{jj}(k)] + O(t^2), \quad j = 2,3,4. \end{aligned}} \tag{4}$$

The leading coefficients are strictly positive for positive transverse weights. At isotropy and $s = 1/2$ this becomes

$$\delta_t^{1j} = \frac{C_A t}{48}\int_{\mathcal{B}_3}\frac{\sum_{k\neq j}(2 - 2\cos k_k)}{8 + Q} + O(t^2). \tag{5}$$

Proof. Use a constant commuting cut flux $X_{1j} = X$, all other cut fluxes zero, and choose the geodesically interpolated mid-connection trivial (links e^{-sX} and $e^{(1-s)X}$; a constant twist is removed in the bulk and only feeds the winding term). The classical defect vanishes on this test. Put

$$Q_Z = \frac{\mathrm{ad}\, Z}{2}\coth\frac{\mathrm{ad}\, Z}{2}.$$

The bridge Hessian, divided by its value at $X = 0$, is

$$\begin{aligned} h_{s,X} &= (1-s)Q_{sX} + sQ_{(1-s)X} \\ &= I + \frac{\sigma}{12}(\mathrm{ad}\, X)^2 + O(|X|^4). \end{aligned}$$

This follows by taking the Hessians of the two squared distances, weighted by $1/(s\tau_{1j})$ and $1/((1-s)\tau_{1j})$. Normalization subtracts the determinant of the unconstrained bridge. Hence the quadratic amplitude is

$$\frac{\sigma}{24}\int_{\mathcal{B}_3}(M_j G_{jj} - 1)\,\mathrm{tr}_{\mathrm{adj}}(\mathrm{ad}\, X)^2 = \frac{C_A\sigma|X|^2}{24}\int_{\mathcal{B}_3}(1 - M_j G_{jj}).$$

The matrix $M^{1/2}GM^{1/2}$ lies between zero and I, with a strict diagonal inequality on a set of positive momentum measure. For (5), invert $4I + C^*C/2$ to get $G = \frac{1}{4}[I - C^*C/(8+Q)]$. □

Suppressing direction 4 and taking $C_A = 2$ gives exactly the cut coefficient $t\int q/R\,/24$ of the D=3 note. The full colour trace fixes this normalization; the isotropic average of Proposition 6 would give a different constant.

3. N3: an explicit integral for each transverse plane

Take identical adjacent layers with a covariantly constant Abelian flux $X_{jk} = BT_3$ in one transverse plane and zero flux in the other two. Here $|T_3| = 1$ for $SU(2)$; j, k, h are the three distinct transverse directions. The interpolated field is an exact stationary point. For equal endpoints, the two zero-image heat-kernel amplitudes cancel the Haar Jacobian, at every s:

$$k_{st_e}(e^{\xi})k_{(1-s)t_e}(e^{-\xi})\,dm \;\propto\; e^{-|\xi|^2/(2\sigma t_e)}\,d\xi.$$

Thus the bridges contribute only M to this determinant. The remaining mid-face amplitude is $-B^2/24$ per (jk) face for $SU(2)$.

Here is a fully specified matrix-integral expression retaining all three edge components. It avoids a scalar Schur reduction of the third component and leaves the lattice integral unevaluated. Put

$$p = k_j, \quad z = k_k, \quad w = k_h, \quad q(v) = 2 - 2\cos v, \qquad d_B = \frac{B/2}{\sin(B/2)}, \quad h_B = d_B\cos(B/2).$$

Use magnetic Weyl symbols, with $[z, p] = iB$. The Hermitian matrix K_B^{jk}, supported on rows and columns j, k, has entries

$$\begin{aligned}
(K_B^{jk})_{jj} &= 2h_B - 2d_B\cos(z + B/2),\\
(K_B^{jk})_{kk} &= 2h_B - 2d_B\cos(p - B/2),\\
(K_B^{jk})_{jk} &= d_B\{(e^{-iB/2} - e^{-iz})e^{ip} + e^{-i(z+B/2)} - e^{iB/2}\},\\
(K_B^{jk})_{kj} &= \overline{(K_B^{jk})_{jk}}.
\end{aligned} \tag{6}$$

The other two plane matrices have the following nonzero entries, with their reverse entries given by complex conjugation:

$$\begin{array}{lll}
(K^{jh})_{jj} = q(w), & (K^{jh})_{hh} = q(p), & (K^{jh})_{jh} = (1 - e^{-iw})(e^{ip} - 1),\\
(K^{kh})_{kk} = q(w), & (K^{kh})_{hh} = q(z), & (K^{kh})_{kh} = (1 - e^{-iw})(e^{iz} - 1).
\end{array} \tag{7}$$

Define

$$H_B = M + W_{jk}K_B^{jk} + W_{jh}K^{jh} + W_{kh}K^{kh} = H_0 + BH_1 + B^2H_2 + O(B^3),$$
$$H_1 = \partial_B H_B|_0\,, \qquad H_2 = \tfrac{1}{2}\partial_B^2 H_B\Big|_0\,.$$

The derivatives here act on the explicit symbols (6)–(7). To include the orbital contribution, use the ordered matrix products

$$\{A, B\} = A_zB_p - A_pB_z,$$
$$\mathcal{Q}(A, B) = A_{zz}B_{pp} - 2A_{zp}B_{zp} + A_{pp}B_{zz}.$$

For a real spectral parameter $v \geq 0$ set

$$\begin{aligned}
R_0 &= (H_0 + vI)^{-1},\\
R_1 &= -R_0\left(H_1R_0 + \frac{i}{2}\{H_0, R_0\}\right),\\
R_2 &= -R_0\left(H_2R_0 + H_1R_1 + \frac{i}{2}\{H_0, R_1\} + \frac{i}{2}\{H_1, R_0\} - \frac{1}{8}\mathcal{Q}(H_0, R_0)\right),\\
\mathcal{I}_{jk}(M, W) &= -\operatorname{Re}\int_{[-\pi,\pi]^3}\frac{d^3k}{(2\pi)^3}\int_0^\infty \operatorname{tr}_3 R_2(k, v)\,dv.
\end{aligned} \tag{8}$$

Equations (6)–(8) are an explicit Brillouin-zone integral with an elementary resolvent parameter. They require only inverses and derivatives of specified three-by-three matrices. Bound (3) guarantees convergence at $v = 0$; $R_2 = O(v^{-2})$ guarantees convergence at infinity.

Derivation. The squared-distance plaquette Hessian, for transported oriented edge fluctuations η_i and background logarithm Y, is

$$L^{(2)} = \frac{1}{2}\left\langle \sum_i \eta_i, Q_Y \sum_i \eta_i \right\rangle + \frac{1}{2}\sum_{i<l}\langle Y, [\eta_i, \eta_l]\rangle, \qquad L = \tfrac{1}{2}|\log m_{\partial g}|^2. \tag{9}$$

In the gauge $U_j = U_h = 1$, $U_k(x) = e^{Bx_jT_3}$, the (jk) operator is equation (11) of the mid-plane note. A term $f(z)E_j$ has Weyl symbol $f(z - B/2)e^{ip}$, giving (6). The zero-flux (jh) and (kh) plaquettes have the covariant curls $(1 - e^{iw})\xi_j + (E_j - 1)\xi_h$ and $(1 - e^{iw})\xi_k + (e^{iz} - 1)\xi_h$, which give (7). This retains the shared h-edge, the radial term Q_Y, and the ordered commutator in (9).

Expanding $(H_B + v) \star (R_0 + BR_1 + B^2R_2) = I$ with $A \star B = AB + (iB/2)\{A, B\} - (B^2/8)\mathcal{Q}(A, B) + O(B^3)$ gives (8). The logarithm identity $\log H = \int_0^\infty [(1+v)^{-1}I - (H+v)^{-1}]\,dv$ then proves that $\mathcal{I}_{jk}$ is the B^2 coefficient of the charged determinant per site. Two charged real colours contribute $\frac{1}{2}\operatorname{Tr}_{\mathbb{R}} \log H = \operatorname{Tr}_{\mathbb{C}} \log H$. The neutral colour is constant. This is a bulk magnetic expansion, taken before imposing quantized flux on a finite torus.

Theorem 1(b). The other three plane shifts are

$$\boxed{\begin{aligned} A_{jk} &= -\frac{C_A}{48} + \frac{C_A}{2}\mathcal{I}_{jk}(M, W), \\ \delta_t^{jk} &= C_A t\tau_{jk}\left(-\frac{1}{24} + \mathcal{I}_{jk}(M, W)\right) + O(t^2). \end{aligned}} \tag{10}$$

For $SU(2)$, the Jacobian contribution is $-1/24$ in A_{jk} and the determinant contribution is exactly (8). For $SU(N)$ the root planes have charges $\alpha(T)$, with $\sum_{\alpha>0}\alpha(T)^2 = C_A|T|^2/2$; this multiplies the unit-charge determinant by $C_A/2$. Also $\log j(X) = \operatorname{tr}(\operatorname{ad} X)^2/24 + O(|X|^4)$, so the heat-kernel amplitude contributes $\frac{1}{2}\log j(X) = -C_A|X|^2/48$. The old-face heat-kernel ratio cancels its Jacobian. Together these give (10). □

As a dimensional reduction check, set $W_{jh} = W_{kh} = 0$. The third component becomes a constant determinant; with $M_j = M_k = 4$ and $W_{jk} = 1/2$, (8) is the B^2 coefficient of $\log\det(8 + K)$ of the D=3 note. Its scalar reduction is exactly that note's integral (2), including the Weyl-product correction. Simply adding three independent copies of that two-dimensional integral would discard the shared fluctuations in (7).

4. The infrared and the size clause

For every fixed positive anisotropy and $s \in (0, 1)$, (3) makes the integrands in (4) and (8) analytic near $k = 0$. Their integral over a ball $|k| < \varepsilon$ is $O(\varepsilon^3)$. The cut integrand even vanishes at $k = 0$. Thus the elementary mid-space propagator supplies **zero coefficient of an infrared logarithmic divergence**. A finite value containing $\log 2$ could still arise from integration over the whole Brillouin zone; its presence would require a matching argument before identifying it with the beta function.

The finite-torus issue of $\mathrm{P}(\alpha)$ survives in $D = 4$. At isotropy and $s = 1/2$, take equal flat $SU(2)$ layers with adjoint twists θ_j and $V_3 = \prod_j N_j$. Since the charged determinant has two transverse eigenvalues, its one-loop winding term is

$$\begin{aligned} \Omega_N(\theta) &= 2\sum_k \log\frac{14 - 2\sum_{j=2}^4 \cos(k_j + \theta_j/N_j)}{14 - 2\sum_{j=2}^4 \cos k_j}, \\ 0 \le \Omega_N(\theta) &\le \frac{7V_3}{n_*}\left(\frac{3}{7}\right)^{n_*}, \qquad n_* = \min_j N_j. \end{aligned} \tag{11}$$

The closed-walk expansion of $\log(14 - \mathsf{A}_\theta)$ proves the bound: at most 6^n walks of length n, winding requires $n \geq n_*$, and $1 - \cos(w \cdot \theta) \leq 2$. The leading factor 2 counts the transverse eigenvalues. At fixed N this term persists as $t \to 0$. The size clause $n_* \geq c_0/t$ gives density at most $(7t/c_0)e^{-c_0 \log(7/3)/t}$. It controls this winding contribution; it leaves the bulk bridge mass intact. Positive bounded anisotropies have analogous exponential decay with ratio-dependent constants.

Theorem 1 uses bulk coefficients, or a volume limit eliminating these winding terms. Extending it to a normalized small-field estimate needs the saddle, connected-kernel, heat-kernel-image and remainder controls specified in §7 of the mid-plane note. P(α)'s size clause alone supplies none of those missing bounds. Moreover a bound with $\alpha < 1$ can absorb an $O(t)$ relative coupling change, so the inequality alone cannot determine the one-loop coefficient.

5. Four halvings: what actually composes

Let $\mathcal{L}_r$ have its first r directions halved, starting with spacing $2a$ in every direction. Then $\mathcal{L}_4$ has spacing a, and the unperturbed heat-time ratios are

r	τ_{12}	τ_{13}	τ_{14}	τ_{23}	τ_{24}	τ_{34}
0	1	1	1	1	1	1
1	1/2	1/2	1/2	2	2	2
2	1/4	1	1	1	1	4
3	1/2	1/2	2	1/2	2	2
4	1	1	1	1	1	1

Equations (4) and (10), relabeling direction 1 as direction r, give the unperturbed defect from $\mathcal{L}_r$ to $\mathcal{L}_{r-1}$ using row $r-1$. The coarsening integration proceeds in reverse order, $r = 4, 3, 2, 1$. Denote the reference heat-kernel law on $\mathcal{L}_r$ by μ_r, its unperturbed defect by $\mathcal{D}_r$, and the accumulated additional action by $\mathcal{R}_r$. Up to vacuum constants, exact conditional integration says

$$\boxed{\mathcal{R}_{r-1}(U) = \mathcal{D}_r(U) - \log E_{\mu_r}\left[e^{-\mathcal{R}_r} \mid p_r U_r = U\right], \qquad \mathcal{R}_4 = 0.} \tag{12}$$

The conditional law here includes all mid-face weights. Proposition 1 specifies it through the bridge integral. Already $\mathcal{D}_4$ has the nonzero Gaussian defect, its nonabelian classical completion and its one-loop amplitude. Their conditional average in the next step changes both the saddle and its Hessian. On smooth fields the Gaussian defect starts at dimension six, but internal momenta in (8) range over the whole Brillouin zone: that smooth-field suppression gives no permission to discard it in a one-loop contraction.

At quadratic fluctuation order, exact composition has the familiar Schur-complement identity

$$\det \begin{pmatrix} A & B \\ B^* & D \end{pmatrix} = \det D \, \det(A - BD^{-1}B^*). \tag{13}$$

It is the **successively updated** determinants that telescope. Replacing each Schur complement by a new nearest-plaquette heat-kernel Hessian changes the finite coupling matching. The four pristine Ψ integrals locate the generated terms; (12) governs their composition. This is the precise qualification needed for the proposed four-factor calculation in §5 of the series/parallel note.

6. Conditional matching theorem and the universal coefficient

Theorem 2 (formal one-loop matching). Suppose a gauge-covariant blocking calculation retains the full generated action as in (12). Assume its perturbative actions have local kernels with controlled small-momentum expansions, the continuum Yang–Mills kinetic normalization,

no additional massless modes, and background Ward identities. Assume that, after a common infrared regulator is inserted, the one-loop determinants match the continuum operators

$$\Delta_{1,\mu\nu} = -D^2\delta_{\mu\nu} - 2\,\mathrm{ad}\,F_{\mu\nu}, \qquad \Delta_0 = -D^2, \qquad \Gamma_1 = \tfrac{1}{2}\log\det\Delta_1 - \log\det\Delta_0, \tag{14}$$

up to infrared-integrable lattice differences and finite local F^2 matching terms. The scalar determinant is the ghost determinant in this background gauge. Its counterpart in a gauge-fixed bridge calculation must be accounted for by the measure and gauge reduction. Assume also that the coupling is extracted using this same background normalization at the two endpoints.

Let $\ell = (\prod_\mu a_\mu)^{1/4}$, let $\zeta_\mu = a_\mu/\ell$ specify the anisotropy, and let $\mathcal{K}$ specify all remaining dimensionless action kernels. Define the finite matching coefficient c by

$$u_R(\mu) = u(\ell) + 2b_0\log(\ell\mu) + c(\zeta,\mathcal{K}) + O(t), \qquad u = g^{-2}.$$

Then one isotropic coarsening gives

$$\boxed{u_{\mathrm{out}}(2a) - u_{\mathrm{in}}(a) = -2b_0\log 2 + c(\zeta_{\mathrm{in}},\mathcal{K}_{\mathrm{in}}) - c(\zeta_{\mathrm{out}},\mathcal{K}_{\mathrm{out}}) + O(t), \quad b_0 = \frac{11N}{48\pi^2}.} \tag{15}$$

If the dimensionless action family returns to itself through one-loop matching accuracy, or if its finite matching terms are subtracted in defining the coupling, (15) gives (1).

Analytical coefficient. Write $F^2 = \sum_{a,\mu,\nu}(F^a_{\mu\nu})^2$, with both orders of μ,ν included, so the classical action is $u\int F^2/4$. The flat-space heat coefficient is

$$a_4(\Delta) = \frac{1}{(4\pi)^2}\int \mathrm{tr}\left(\tfrac{1}{2}E^2 + \tfrac{1}{12}\Omega_{\mu\nu}\Omega_{\mu\nu}\right).$$

For (14), the traces are $4C_AF^2$ for the vector E^2, $-4C_AF^2$ for its Ω^2, and $-C_AF^2$ for the ghost Ω^2. Consequently

$$\tfrac{1}{2}a_4(\Delta_1) - a_4(\Delta_0) = \frac{C_A}{16\pi^2}\left(\frac{5}{6} + \frac{1}{12}\right)\int F^2 = \frac{11C_A}{192\pi^2}\int F^2.$$

The proper-time shell $a^2 < \rho < (2a)^2$ gives $\Gamma_{1,\mathrm{shell}} = -(11C_A/192\pi^2)\log 4\int F^2$. Multiplication by 4 to extract u proves $\Delta u = -11C_A\log 2/(24\pi^2)$. This normalization calculation uses Vassilevich, §4.2.1, equations (4.28)–(4.34) (passage; DOI 10.1016/j.physrep.2003.09.002, metadata checked). It is the established one-loop coefficient.

Telescoping. Each reverse directional step multiplies ℓ by $2^{1/4}$, and its universal scalar part is $-(2b_0/4)\log 2$ in this choice of scale. Its finite part is $c_{\mathrm{before}} - c_{\mathrm{after}}$. Summing the four steps cancels the three intermediate matching terms and gives (15). Plane-dependent anisotropy counterterms obey the same endpoint cancellation in a fixed background convention. Splitting c into a reference anisotropy function and an action-dependent remainder is a scheme choice. The anisotropy function cancels because $\zeta_{\mathrm{in}} = \zeta_{\mathrm{out}}$; the remaining endpoint difference requires $\mathcal{K}$ as well. Equation (12) changes $\mathcal{K}$ even when the table returns to row 0. □

Many-step telescoping. Coarsen a very fine lattice n times, each step's output being the next step's input. Summing (15), the intermediate matching terms cancel as in the four-step telescoping:

$$u_n - u_0 = -2b_0\,n\log 2 + c(\zeta_0,\mathcal{K}_0) - c(\zeta_n,\mathcal{K}_n) + \sum_{k<n} O(t_k).$$

Along the coarsening the couplings grow from a small initial value, and if the final coupling is still small, $t_k \approx 1/(2b_0(n-k)\log 2 + 1/t_n)$ and $\sum_k O(t_k) = O(\log n)$. The per-step average therefore tends to $-2b_0\log 2$ as soon as $c(\zeta_n,\mathcal{K}_n) = o(n)$, that is, as soon as the generated action

shapes stay in a family on which the finite matching coefficient is bounded. A stationary family gives (1) at every step; boundedness along the iteration is enough for the universal rate on average. Proving it still requires control of the generated actions through (12).

For the pristine heat-kernel family, matching two isotropic bare regularizations along a line of fixed renormalized coupling has the same endpoint c and gives (1). Identifying the coupling obtained by projecting a single blocked density onto its plaquette term with that bare-family coupling requires the finite conversion in (15). This is the same distinction behind the finite Λ ratios in the zero-spacing note: if $u_B = u_A + d + O(t)$ at the same scale, then $\Lambda_B/\Lambda_A = e^{-d/(2b_0)}$ at one loop.

7. The integral carrying the logarithm, and general cuts

The infrared singularity can be isolated without numerical quadrature. With $\widehat{k}^2 = \sum_{\mu=1}^{4}(2 - 2\cos k_\mu)$, use the reference integral

$$J(a\mu) = \int_{[-\pi,\pi]^4} \frac{d^4k}{(2\pi)^4} \frac{1}{(\widehat{k}^2 + (a\mu)^2)^2} = \frac{1}{8\pi^2} \log \frac{1}{a\mu} + j_{\mathrm{lat}} + o(1). \tag{16}$$

To see its residue, split at $a\mu \ll \varepsilon \ll 1$. In the inner ball $\widehat{k}^2 = k^2 + O(k^4)$; the correction is integrable. The spherical measure is $2\pi^2 r^3 dr/(2\pi)^4$, giving $\log(\varepsilon/(a\mu))/(8\pi^2)$. The outer region has a finite limit. Equivalently, integrating one continuum momentum first gives

$$\int_{\mathbb{R}} \frac{dk_1}{2\pi} \frac{1}{(k_1^2 + r^2)^2} = \frac{1}{4r^3}, \qquad \int_{\mu a < |k_\perp| < \varepsilon} \frac{d^3k_\perp}{(2\pi)^3} \frac{1}{4|k_\perp|^3} = \frac{1}{8\pi^2} \log \frac{\varepsilon}{a\mu}. \tag{17}$$

This is the three-dimensional propagator integral whose infrared end carries the logarithm **after matching to the full massless four-dimensional determinant**. The conditional propagator G of (2) has a different infrared end, proved regular in §4. Successive Schur complements and the retained-field determinant must reconstruct the massless expression before (17) applies. Thus assigning the logarithm to the infrared end of each elementary massive Ψ sum would miss the required reassembly.

Under Theorem 2's hypotheses, the singular contribution to the inverse coupling is $-(11C_A/3)J$, as fixed by (14). Therefore

$$-\frac{11C_A}{3}[J(a\mu) - J(2a\mu)] \longrightarrow -\frac{11C_A}{24\pi^2} \log 2. \tag{18}$$

Changing a local regularization with the same continuum kinetic operator changes j_{lat} and the finite tensor integrals, while the residue in (16) stays fixed. In the inner ball the difference is integrable; in the outer region it contributes only to c. The scheme-independent object is this logarithmic residue. The finite values of (8) depend on M, W and s. Evaluating them numerically would compute one-step finite shifts, and would still require (12) and the endpoint matching to establish a universal isotropic step. The lattice background-field method and its symmetry requirements are established in Lüscher and Weisz (1995) (abstract; their renormalization result supplies context, with transfer to this blocking retained as an explicit hypothesis).

General fractions. Equations (2), (4) and (8)–(10) already give the one-step answer for every fixed $s \in (0,1)$. In (8) the fraction enters only through $M_j = 1/(\sigma\tau_{1j})$; the explicit N2 prefactor is $\sigma/24$. The coefficients are invariant under $s \leftrightarrow 1-s$ and their finite values generally depend on s. The three-dimensional Jacobian term in (10) also remains present as s approaches an endpoint.

A cut produces both lengths sa_1 and $(1-s)a_1$. Repeating such cuts creates a nonuniform lattice, so the resulting global scale change needs a specified schedule and matching convention. A factor $\log(1/s)$ alone would describe following just one daughter; it omits the other daughter and its dual-volume weights. For shape-regular schedules whose successive effective actions

satisfy Theorem 2 and whose endpoint schemes agree, the coefficient per logarithm of the physical scale is $2b_0$, independent of cut positions. Finite matching terms and winding estimates can depend on the schedule. Uniform control for fractions tending to 0 or 1 and for arbitrary nonuniform meshes remains an additional obligation.

8. Consequence for STATE

The formal anisotropic one-step coefficients, exact generated-action recursion (12) and conditional matching identity (15) retain their distinct assumptions. Finite endpoint matching for one pristine step and the repeated-step norm criteria are developed in composition. They do not give depth-uniform control of the interacting generated action or a continuum theory with a physical spectral gap. New research remains paused.